Machine Learning Project

AI1215 — Introduction to Machine Learning

Course Project — Summer 2026

1. Introduction

Machine learning is only useful when it solves a real problem for real people. In this project, you will go through the full pipeline of applied ML: find a meaningful problem, collect raw data, build and evaluate a model, and present your work.

You are free to choose **any topic that matters** — something you are already researching, a problem you encounter in daily life, or one of the suggestions in Section 3. The only requirements are that (1) the problem is genuinely useful to someone, (2) you collect, scrape, or download the data, and (3) you analyze it and train machine learning models to address the problem.

2. What You Will Build

Every project must include the following components:

1. **Data collection** — gather raw data from the web, an API, a sensor, or a public database. Ready-to-use datasets (Kaggle, UCI) are also acceptable.
2. **Exploratory Data Analysis (EDA)** — understand your data before modelling: distributions, correlations, missing values, outliers.
3. **Preprocessing & Feature Engineering** — clean, transform, and enrich the data to make it model-ready.
4. **Model Training & Selection** — train different model types, compare them, and justify your final choice.
5. **Evaluation & Analysis** — measure performance with an appropriate metric, discuss errors, and draw conclusions about what the model learned.
6. **Written Report** — a concise document describing your methodology and findings.
7. **Code Submission** — clean, reproducible code.
8. **Oral Presentation** — present your work and answer questions on both your project and course content.

3. Suggested Topics

Below are example topics with suggested data sources and ML tasks. You are **not** required to pick one of these — they are starting points. You may also propose your own topic (see Section 4).

Medical & Health

Hospital Readmission Prediction *Classification*

Predict whether a patient will be readmitted within 30 days of discharge. Use the **UCI Diabetes 130-US Hospitals** dataset (free, direct download) or apply for access to **MIMIC-IV** (`physionet.org`) — free for academic use but requires registration and a data-use agreement (allow 1–2 weeks).

Who benefits: hospitals and insurers reducing avoidable readmissions.

Air Quality & Health Risk Forecasting *Regression / Classification*

Collect historical AQI (Air Quality Index) readings via the **OpenWeatherMap API** or **IQAir API**, combine with UAE National Centre of Meteorology weather data, and predict next-day pollution levels or classify risk tiers (Good / Moderate / Unhealthy).

Who benefits: residents and city planners managing outdoor activity decisions.

Mental Health Sentiment Analysis *Classification*

Collect public posts from Reddit communities (r/depression, r/anxiety) using the official **Reddit PRAW API** (free, requires registration) and classify posts by distress level. Alternatively, use the **CLPsych 2015** shared-task dataset for distress classification.

Note: this topic involves sensitive content — do not attempt to identify individual users.

Who benefits: mental health platforms triaging users who need urgent support.

Financial & Economic

Real Estate Price Prediction (UAE) *Regression*

Download official transaction records from the **Dubai Land Department open data portal** (`dubailand.gov.ae`) or the **UAE Government Open Data** portal (`data.gov.ae`). Features: property type, area (sqft), location, floor, transaction date. Predict sale price per square foot.

Who benefits: buyers, sellers, and real estate agents seeking fair price estimates.

Used Car Price Estimation *Regression*

Use one of the several publicly available used-car datasets on **Kaggle**, such as the *Vehicle Dataset from CarDekho*, the *UK Used Car Dataset*, or the *Used Car Price Prediction* dataset. Features: make, model, year, mileage, condition, transmission. Predict market value.

Who benefits: private sellers and buyers avoiding over/under-pricing.

Financial News Sentiment & Stock Movement *Classification*
Collect financial news headlines via the **NewsAPI** (free developer tier) or the **GDELT Project** dataset (`gdeltproject.org`, freely downloadable), and align them with daily price changes from **Yahoo Finance** (`yfinance`). Predict whether a stock goes up or down the next day.
Who benefits: retail investors and algorithmic trading systems.

Science & Environment

Crop Yield Prediction *Regression*
Download agricultural data from the **FAO** database (`fao.org/faostat`) combined with climate data from **NASA POWER** or **WorldClim**. Predict crop yield (tonnes/hectare) from temperature, rainfall, soil quality, and fertilizer use.
Who benefits: farmers and food-security agencies planning resource allocation.

Energy Consumption Forecasting *Regression*
Use **DEWA open data** or the **UCI Individual Household Power Consumption** dataset. Supplement with UAE weather data. Forecast next-hour or next-day electricity demand.
Who benefits: grid operators reducing over-generation and blackout risk.

Business & Society

Job Salary Prediction *Regression*
Use the **Glassdoor Jobs & Salaries** dataset or the **Data Science Job Salaries** dataset (both freely available on Kaggle), or query the **US Bureau of Labor Statistics API** (`bls.gov/developers`) for occupational wage data. Features: job title, experience level, education, location, industry. Predict salary range.
Who benefits: job seekers negotiating fair compensation.

Restaurant Rating Prediction *Regression / Classification*
Collect restaurant data via the **Google Places API** or the **Yelp Fusion API** (both offer free developer tiers). Features: cuisine type, location, price range, number of reviews, opening hours. Predict star rating or success category.
Who benefits: new restaurant owners and investors assessing viability.

Customer Churn Prediction *Classification*
Use the **IBM Telco Customer Churn** dataset (available on Kaggle) or the **Cell2Cell Telecom Churn** dataset. Features: tenure, monthly charges, contract type, usage

patterns, support calls. Predict which customers are at risk of leaving.
Who benefits: subscription businesses targeting retention efforts.

4. Custom Topic

If none of the above fit your interests, you may propose your own topic. A valid custom topic must:

- Address a **real need** — identify clearly who will benefit and how.
- Have a **collectable dataset** — describe where the data will come from and how you will obtain it *before* the topic is approved.
- Be suitable for a **supervised ML task** — classification or regression.

Submit a one-paragraph topic proposal (topic name, problem statement, data source, ML task) **before beginning data collection**. You will receive approval or feedback within 48 hours.

5. Data Collection Guidelines

Below are recommended tools and approaches for obtaining data, whether by scraping, calling an API, or downloading from a public source.

Method	**Tools**	**Typical use case**
Web scraping	`requests`, `BeautifulSoup`, `Selenium`	Product listings, news articles, property data
REST APIs	`requests` + API key	Weather, finance, social media, maps
LLM-assisted labelling	Claude / GPT API	Generating labels or features when ground truth is unavailable
Public databases	Direct download (FAO, WHO, World Bank, UAE open data)	Structured national/global statistics
Surveys	Google Forms, direct collection	When no digital source exists

Minimum dataset size: at least **500 samples** with at least **5 meaningful features**. For scraping-based projects, submit the raw scraped file alongside your cleaned version.

Note on AI assistance: You may use Claude, GPT, or similar tools to help label data, generate synthetic augmentation, or assist with code — but you must disclose this in your report and explain exactly how it was used.

6. Team Formation and Grading

6.1 Teams

Teams may consist of **1 to 4 students**. Larger teams are expected to produce more thorough analysis — graders will take team size into account when evaluating scope.

6.2 Grading Breakdown (50% of module grade)

Component	Project Weight	Module Grade
Oral Presentation	60%	30%
Written Report	20%	10%
Code & Results	20%	10%
Total		**50%**

6.3 Code & Results (10%)

Requirements:

- Code must be reproducible: the teaching team must be able to run it end-to-end and obtain your reported results.
- Include the raw data file(s) or a script that re-collects them.

Grading:

- **Results (50%):** Model outperforms a simple baseline (e.g., always predicting the majority class or the mean). Report your chosen metric clearly.
- **Code quality (50%):** Clean, well-structured, runs without errors, with a clear `README`.

Submit on Moodle a ZIP file named `TeamName_Code.zip` containing:

1. `code/` — all notebooks or scripts
2. `data/` — raw and processed data files (or a collection script)
3. `README.txt` — instructions to reproduce your results
4. `requirements.txt` — Python packages and versions

7. Written Report (10%)

Format: Up to 7 pages (excluding cover page and references), PDF.
Cover page must include: team name, student names, project title, and chosen topic.

Section 1: Problem & Data (25%)

- Describe the problem and who it helps.
- How did you collect the data? Include source URLs or API endpoints.
- Key statistics: number of samples, features, class balance (if classification), missing values.
- At least one visualization of the data distribution.

Section 2: Preprocessing & Feature Engineering (25%)

- How you handled missing values and categorical features, and why.
- Any new features you engineered and the intuition behind them.
- Train/validation/test split strategy.

Section 3: Model Selection & Results (35%)

- At least two model types compared in a table (model, metric on validation set).
- Final model choice and justification.
- Error analysis: where does the model fail? What might explain these failures?

Section 4: Conclusions (15%)

- What did the model learn? Are the results useful in practice?
- Limitations and potential improvements.

8. Oral Presentation (30%)

Schedule: Week 7 (final week) **Duration:** 15 minutes + 10 minutes Q&A
All team members must be present and must speak.

Suggested structure:

1. **Problem & Motivation** (2 min) — what problem you tackled and why it matters
2. **Data Collection** (2 min) — how you collected data, challenges encountered

3. **Methodology** (4 min) — preprocessing, models tried, evaluation
4. **Results & Insights** (2 min) — what worked, what did not
5. **Demo (optional)** (2 min) — live prediction on a new example

Q&A note: Questions will cover **both your project and all course content**. Every team member may be asked independently, so everyone must understand the full project and the course material.

Grading: Content 50% · Presentation skills 30% · Q&A 20%

9. Academic Integrity

- You may use online resources and AI tools, but cite all sources in your report.
- Code and report must be your own — no copying from other teams.
- Respect the terms of service of any website you scrape. Do not overload servers; add delays between requests.

10. Useful Libraries

Library	Use
`requests`, `BeautifulSoup`	Web scraping
`selenium`	Scraping JavaScript-heavy pages
`pandas`, `numpy`	Data manipulation
`matplotlib`, `seaborn`	Visualization
`scikit-learn`	Preprocessing, models, evaluation
`xgboost`, `lightgbm`	Gradient boosting
`transformers` (HuggingFace)	Text / NLP tasks
`yfinance`	Financial data
`anthropic`, `openai`	LLM-assisted labelling

11. Timeline

Deadline	Deliverable
Week 2	Submit team and topic
Week 6	Submit report and code
Week 7 (Final week)	Present

12. Final Checklist

- Topic proposal submitted and approved before data collection begins
- Raw data collected and included in submission
- At least two models trained and compared
- Baseline clearly defined and beaten
- Report is PDF, up to 7 pages, all four sections present
- Code runs from scratch without errors
- `README.txt` explains how to reproduce results
- All team members have a speaking part prepared
- Q&A preparation covers both the project and course content

Pick a problem that excites you — the best projects come from genuine curiosity.